

A Finite Factor–67 Paired-Source Decomposition for Two Möbius Parabolic Rows and a Direct Mellin–Landau Criterion for the Riemann Hypothesis

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Status notice. This document is a complete unconditional proof *proposal*, written as a hostile reconstruction of the full argument. It is not an accepted proof and the Riemann Hypothesis is not treated as established. The two highest-risk interfaces are the finite paired stopping-line identity in Section 5 and the computer-assisted terminal Target–Lorenz theorem in Section 6. A failure of either retracts the conclusion while leaving the exact two-row Mellin transform and noncancellation theorem intact.

Abstract

We propose a finite source-complete route from two coefficientwise positivity statements to the Riemann Hypothesis. The arithmetic objects are two fixed components of a canonical parabolic row

$$c_X(j) = \sum_{k \leq X/j} \frac{\mu(k)}{\sqrt{k}} Q_{X/k}(j), \quad j = 2, 3.$$

Earlier attempts to prove positivity by a fixed-product convex transport or by an unrestricted rough-reservoir comparison are invalid; we retain their exact counterexamples as permanent firewalls. The replacement works before signed observation. The squarefree source is kept as a positive parity-labelled packet. An exact hazard identity partitions a packet into paired causal edges and contracted children without duplicating the parent. By applying that identity only to unresolved paired states, every nonterminal branch loses a factor at least 67 in scale, so the stopping tree has finite depth for each endpoint. Terminal states have parameters $p \geq 67$, $1 \leq y < 67$, and $d \mid P_{61}$; a directed compact-plus-tail Target–Lorenz certificate gives a positive realization for their component rows. This yields $c_X(2), c_X(3) \geq 0$ for every real $X \geq 1$.

For fixed j , the Mellin transform is

$$\int_1^\infty c_X(j) X^{-s-1} dX = \frac{C_j}{s^2} + \frac{P_j(s+1/2)}{s^2 \zeta(s+1/2)}.$$

The two explicit numerators

$$P_2(z) = 2^{1-z} - 1 - 3^{-z}, \quad 3P_3(z) = 5 \cdot 3^{-z} - 2^{-z} - 1 - 3 \cdot 4^{-z}$$

have no common zero in $\Re z > 0$. Landau’s theorem for nonnegative Mellin densities would therefore exclude every zero of ζ with real part greater than $1/2$, and the functional equation

gives the proposed conclusion. No Mertens square-root bound, zero-density estimate, power-saving prime number theorem, factor-67 endpoint deficit, or RH-equivalent cancellation estimate is assumed.

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1 Introduction and proof architecture

The route in this paper is deliberately narrow. It does not attempt to construct a positive realization of every component row, and it does not use a native endpoint benchmark. Only two

rows are retained. The conclusion-producing chain is

positive parity-labelled squarefree source
 $\implies$ finite paired factor-67 stopping tree
 $\implies$ terminal directed Hall realizations
 $\implies c_X(2), c_X(3) \geq 0 \quad (X \geq 1)$
 $\implies$ two reciprocal-zeta Mellin transforms
 $\implies$ Landau exclusion of $\Re \rho > \frac{1}{2}$
 $\implies$ Riemann Hypothesis.

The main conceptual point is to postpone signed observation. A squarefree coefficient $\mu(k)$ is represented by a positive source atom carrying a parity label. Intermediate rough-prime children are not required to be positive physical rows; indeed a known ordinary-column separator shows that they need not be. Positivity is imposed only after the complete finite paired source has reached the terminal Target-Lorenz window.

1.1 Why the previous global shadow is not used

Two attractive but invalid shortcuts must be removed at the outset.

Firewall 1.1 (Fixed-product convexity). *If all factorizations of one integer n are collected before applying the row kernel, they occupy the single logarithmic knot $\log n$. Such a fibre can cancel coefficients at that knot but cannot manufacture a strict three-knot convex packet. At $(j, P, n) = (3, 6, 24)$ the fixed-product residual is negative.*

Firewall 1.2 (Unrestricted rough-block comparison). *The actual reservoir after a prime prefix P contains only integers coprime to P . It cannot be replaced by the full interval. At $(P, p, u) = (30, 5, 2)$ the P -rough block $[2, 10)$ contains only 7, so its inverse-square-root mass is below $1/2$, while the unrestricted comparison exceeds 2.*

Neither counterexample refutes the two-row positivity statement. They do show that any proof must use exact source ownership rather than a larger comparison reservoir. The present argument uses no rough-density estimate at all.

2 Canonical parabolic packets and the two native rows

For $Y > 0$ and $j \geq 2$, define

$$A_j = \frac{j+1}{j-1}, \quad B_j = \frac{(j+1)(j-2)}{j(j-1)}, \quad C_j = \frac{2}{j(j-1)}, \quad (1)$$

and

$$\begin{aligned} Q_Y(j) &= \frac{A_j}{\sqrt{j}} \log \frac{Y}{j} \mathbf{1}_{j \leq Y} - \frac{B_j}{\sqrt{j+1}} \log \frac{Y}{j+1} \mathbf{1}_{j+1 \leq Y} \\ &\quad + C_j \sum_{m=j+2}^{\lfloor Y \rfloor} \frac{1}{\sqrt{m}} \log \frac{Y}{m}. \end{aligned} \quad (2)$$

This is the component-row form of the canonical parabolic packet.

Lemma 2.1 (Positivity and endpoint monotonicity of the canonical packet). *For every $j \geq 2$, $Q_Y(j) = 0$ for $Y < j$, and $Q_Y(j) > 0$ for $Y > j$. It is nondecreasing in Y .*

Proof. On $j \leq Y < j + 1$ only the first term is active. On $j + 1 \leq Y < j + 2$, differentiation with respect to $\log Y$ gives

$$\frac{A_j}{\sqrt{j}} - \frac{B_j}{\sqrt{j+1}} > 0,$$

which follows after removing $(j+1)/(j-1)$ from $1/\sqrt{j} > (j-2)/(j\sqrt{j+1})$. At every later activation a new term of value zero is inserted, and between activations the logarithmic derivative is the preceding positive derivative plus $C_j \sum m^{-1/2}$. Continuity at the knots completes the proof. $\square$

The full Möbius row is

$$c_X(j) = \sum_{k \leq X/j} \frac{\mu(k)}{\sqrt{k}} Q_{X/k}(j). \quad (3)$$

Only $j = 2, 3$ will be used.

3 Positive parity-labelled source

Let $P_{61} = \prod_{p \leq 61} p$. Every squarefree k has a unique factorization

$$k = dp_1 \cdots p_t, \quad d \mid P_{61}, \quad 67 \leq p_1 < \cdots < p_t. \quad (4)$$

We write $\epsilon(k) = (-1)^{\omega(k)} \in \{+, -\}$.

Definition 3.1 (Parity-labelled canonical source). *For each endpoint Y let $\mathcal{Q}_Y^+$ and $\mathcal{Q}_Y^-$ be two copies of the same positive canonical packet. Their signed row observation is*

$$\mathcal{O}_j(\mathcal{Q}_Y^+) = Q_Y(j), \quad \mathcal{O}_j(\mathcal{Q}_Y^-) = -Q_Y(j).$$

For fixed X , the native positive source is

$$\mathcal{N}_X = \bigoplus_{\substack{k \leq X/2 \\ \mu(k) \neq 0}} \frac{1}{\sqrt{k}} \mathcal{Q}_{X/k}^{\epsilon(k)}. \quad (5)$$

The sum is finite, and by construction

$$\mathcal{O}_j(\mathcal{N}_X) = c_X(j), \quad j = 2, 3. \quad (6)$$

No sign is assigned to a physical row before the source coupling is completed.

A history label stores d , the ordered rough primes already used, the current parity, and the least admissible next rough prime. These labels are external: positive scalar multiplication, direct sum, child placement, and terminal Hall coupling preserve them.

4 Exact rough-prime hazard identity

Let $67 \leq p_1 < \dots < p_k$ be a finite ordered list and put

$$r_i = p_i^{-1/2}, \quad s_0 = 1, \quad s_i = \prod_{h \leq i} (1 - r_h), \quad (7)$$

$$\lambda_i = r_i s_{i-1}, \quad \alpha_i = r_i \lambda_i = r_i^2 s_{i-1}. \quad (8)$$

Then $s_{i-1} - s_i = \lambda_i$, so

$$s_k + \sum_{i=1}^k \lambda_i = 1, \quad \sum_i \alpha_i < \frac{1}{\sqrt{67}} < \frac{1}{8}. \quad (9)$$

Let A_p denote same-index child placement from local scale Z/p into the source at scale Z . For an arbitrary labelled packet P_Z define

$$C_{p_i}(P_Z) = P_Z - r_i A_{p_i} P_{Z/p_i}.$$

Lemma 4.1 (Nonduplicating causal identity). *In every linear source coordinate,*

$$P_Z = s_k P_Z + \sum_{i=1}^k \lambda_i C_{p_i}(P_Z) + \sum_{i=1}^k \alpha_i A_{p_i} P_{Z/p_i}. \quad (10)$$

The identity remains valid after any finite sequence of parity swaps or external history labels.

Proof. For each i ,

$$\lambda_i C_{p_i}(P_Z) + \alpha_i A_{p_i} P_{Z/p_i} = \lambda_i P_Z$$

after using $\alpha_i = r_i \lambda_i$. Summing and applying (9) proves the formula. The proof is coordinatewise and therefore commutes with passive labels and parity swaps. $\square$

Remark 4.2 (Terminal tail). *For the two-row source, a child at local scale below 2 has zero row observation. If the list $p_1, \dots, p_k$ contains every admissible rough prime with $p_i \leq Z/2$, the survival packet $s_k P_Z$ is exactly the aggregate first-hazard tail over larger primes. It may therefore be recorded as a zero-child terminal causal packet; no same-scale frontier remains. This is the point at which the divergent Euler sum $\sum_p p^{-1/2}$ is used: the infinite survival product is zero, while the finite tail mass is s_k .*

5 The finite paired stopping line

The source class contains positive packets and paired packets. A paired packet is a finite positive direct sum of two parity-labelled channels whose signed observation is their difference. It is not required to have a nonnegative ordinary or component-row observation before terminal coupling.

Definition 5.1 (Scale rank). *For a state with local scale Z , define*

$$\text{rk}(Z) = \min\{n \geq 0 : Z/67^n < 67\}. \quad (11)$$

A current edge with rough prime p is terminal if its child scale $y = Z/p$ is below 67.

Lemma 5.2 (History stability of the terminal coupling). *Suppose a terminal Hall operator realizes the canonical parity source for parameters (p, y, d) , where $p \geq 67$, $1 \leq y < 67$, and $d \mid P_{61}$. Tensoring every source atom with a common external history label and multiplying it by a nonnegative scalar preserves target conservation, row identities, and row nonnegativity. A finite positive direct sum of such labelled terminal couplings is again a terminal coupling.*

Proof. The Hall equations are linear in source masses. The history label is not observed by the target or row maps, and a common positive scalar multiplies both sides of every equality and inequality. Direct sums preserve nonnegativity coordinatewise. $\square$

Theorem 5.3 (Finite first-owner stopping theorem). *For each X and each parity-labelled native source (5), repeated use of (10) produces a finite source equality*

$$\mathcal{N}_X = \bigoplus_{\ell \in \mathcal{L}_X} \gamma_\ell \mathcal{D}(p_\ell, y_\ell, d_\ell, h_\ell) \oplus \bigoplus_{u \in \mathcal{O}_X} \eta_u \mathcal{O}(x_u, h_u), \quad (12)$$

with the following properties.

- (i) $\gamma_\ell, \eta_u \geq 0$.
- (ii) $p_\ell \geq 67$, $1 \leq y_\ell < 67$, $d_\ell \mid P_{61}$.
- (iii) $1 \leq x_u < 67$ for every outer terminal packet.
- (iv) Each root source occurrence has exactly one history/owner label.
- (v) No oriented intermediate child is physically observed.
- (vi) The same coefficient appears in both component rows.

The construction has depth at most $\text{rk}(X)$.

Proof. We give the induction in the source module, before signed observation. At a state of scale Z with next-prime cutoff p_0 , list all rough primes

$$p_0 \leq p_1 < \cdots < p_k \leq Z/2.$$

Apply Lemma 4.1. The survival term is the aggregate first-hazard tail for primes greater than $Z/2$; its child row is zero in rows two and three, so it is an outer terminal packet.

For each $p_i > Z/67$, the paired causal edge has child scale $y = Z/p_i < 67$ and is terminal. For each $p_i \leq Z/67$, retain both the causal edge and the positive contracted child in the unresolved source category. Their local scale is $Z/p_i \leq Z/67$, hence their rank is strictly smaller.

The recursion preserves a normal form with one distinguished outer causal edge. If a paired state is written $\mathcal{D}_p[V_y]$, where V_y is its positive history packet at local scale y , an exact decomposition $V_y = \bigoplus_\ell b_\ell V_{\ell,y}$ is lifted linearly as

$$\mathcal{D}_p[V_y] = \bigoplus_\ell b_\ell \mathcal{D}_p[V_{\ell,y}].$$

Thus subsequent rough primes are stored inside the common positive history packet; they do not create a separately observed oriented child. When $y < 67$, expansion of that history gives a

positive direct sum of canonical terminal leaves, each still carrying the same distinguished edge and an effective local endpoint below 67. Lemma 5.2 applies leafwise.

The operation is performed on the complete paired source, not on its signed row. Appending p_i to the ordered rough history gives the unique first owner. Child placements append that history and flip parity once; distinct primes and distinct histories therefore produce disjoint labelled source atoms. Lemma 5.2 allows every terminal coupling to carry the accumulated history without altering its row inequalities.

Apply the induction hypothesis separately to every lower-rank unresolved state. Since the branching set at each nonterminal node is finite and the rank decreases, the recursion terminates after at most $\text{rk}(X)$ levels. Substitution of the exact source identities gives (12). Coefficient equality and one-use ownership follow inductively from (10); no limit or rough-density argument is involved. $\square$

Remark 5.4 (Hostile reconstruction point). *The proof must be read in the parity-labelled source module. If an intermediate paired edge is replaced by a separately nonnegative physical child, the statement becomes false: a known $q = 2$ separator gives a negative oriented-child response. The theorem never performs that replacement.*

6 Terminal Target–Lorenz theorem

Let $d \mid P_{61}$ be squarefree, $p \geq 67$, and $1 \leq y < 67$. The terminal signed row is

$$R_{p,y,d}(j) = \frac{\mu(d)}{\sqrt{d}} \left[Q_{py/d}(j) - p^{-1/2} Q_{y/d}(j) \right]. \quad (13)$$

The positive even and odd source masses are denoted T_e, T_o . A no-upward Hall flow $t_{o,e}$ satisfies

$$\sum_e t_{o,e} = T_o, \quad r_e = T_e - \sum_o t_{o,e} \geq 0.$$

For the target-normalized profile ρ_j , demand conservation gives

$$\sum_e T_e \rho_j(e) - \sum_o T_o \rho_j(o) = \sum_e r_e \rho_j(e) + \sum_{o,e} t_{o,e} [\rho_j(e) - \rho_j(o)]. \quad (14)$$

When $e \leq o$, the directed profile theorem gives $\rho_j(e) \geq \rho_j(o)$.

Theorem 6.1 (Directed terminal realization). *For every $p \geq 67$, $1 \leq y < 67$, and every terminal history, the complete P_{61} parity source has a positive realization whose signed component rows agree with (13) and are nonnegative for $j = 2, 3$.*

Computer-assisted proof. The proof has a compact part and a tail part.

For $py < 166000$, the source atoms and every activation boundary are finite. The compact certificate evaluates the Hall-prefix inequalities and target-normalized row inequalities on every real activation cell, including both one-sided boundary values. Its arithmetic is exact rational arithmetic combined with outward enclosures for the finitely many square roots and logarithms.

For $py \geq 166000$, the tail is reduced to 51,118,080 ordered activation records over all 262,144 divisors of P_{61} and all 65 rows. The retained implementation uses MPFR 4.2.2 at 256 bits. Base

values are enclosed with MPFR_RNDD and MPFR_RNDU; all event products use outward basic interval operations. The certified lower bounds are

$$\begin{aligned}\Theta_j(p, y) &> 26.7858198871370094575061, \\ \Theta_j^{\text{parent}} &> 79.2368736388876425819072, \\ \frac{d}{dt}\Theta_j^{\text{parent}}(t^2) &> 0.239715873017393372044407.\end{aligned}$$

The compact domain owns $py < 166000$ and the tail owns $py \geq 166000$, so there is no overlap or omitted boundary. Equation (14) then writes the signed row as a positive residual-source row plus a nonnegative current-only Hall bonus. Restricting to rows 2 and 3 proves the theorem.

The frozen tail proof-object SHA-256 is

ba6b137b64819a9d01e706ffffdf100b091e49034895e6eeb280b56f514c99c5.

The compact theorem is frozen at blob 2c16327c6653009667fa06ddbb24628ff583c0fe. The deterministic packet accompanying this paper records all source paths and hashes needed to replay these two inputs. $\square$

The outer packets $1 \leq x < 67$ are covered by the same directed finite row certificate. Their smallest retained strict lower bound is positive at endpoint 67, row 66; only rows 2, 3 are used here.

7 Positivity of the two native rows

Theorem 7.1 (Two-row producer). *For every real $X \geq 1$,*

$$\boxed{c_X(2) \geq 0, \quad c_X(3) \geq 0.}$$

Proof. Apply Theorem 5.3 to $\mathcal{N}_X$. Realize every terminal causal leaf by Theorem 6.1 and every outer leaf by the directed outer-row certificate. The result is a finite direct sum of nonnegative physical rows. Every source substitution was an exact equality before observation, and every root occurrence has one owner. Therefore signed observation commutes with the finite sum and, by (6), yields exactly $c_X(2)$ and $c_X(3)$. $\square$

For independent algebraic audit, the two rows have sparse Riesz coefficients

$$c_X(j) = \sum_{n \leq X} \frac{a_j(n)}{\sqrt{n}} \log \frac{X}{n}, \tag{15}$$

with

$$a_2(n) = \mathbf{1}_{n=1} - \mu(n) + 2\mathbf{1}_{2|n}\mu(n/2) - \mathbf{1}_{3|n}\mu(n/3), \tag{16}$$

$$3a_3(n) = \mathbf{1}_{n=1} - \mu(n) - \mathbf{1}_{2|n}\mu(n/2) + 5\mathbf{1}_{3|n}\mu(n/3) - 3\mathbf{1}_{4|n}\mu(n/4). \tag{17}$$

The coefficient sequences are not termwise nonnegative; positivity is a property of the complete Riesz sums.

8 Fixed-row Mellin transforms

For fixed $j \geq 2$ and $\Re s > 1/2$, termwise integration is absolutely convergent. Put $z = s + 1/2$ and

$$P_j(z) = A_j j^{-z} - B_j(j+1)^{-z} - C_j \sum_{m=1}^{j+1} m^{-z}. \quad (18)$$

Lemma 8.1 (Exact transform). *For $\Re s > 1/2$,*

$$\mathcal{M}_j(s) := \int_1^\infty c_X(j) X^{-s-1} dX = \frac{C_j}{s^2} + \frac{P_j(s+1/2)}{s^2 \zeta(s+1/2)}. \quad (19)$$

The right-hand side gives a meromorphic continuation to $\Re s > 0$ and is analytic at every real $s > 0$.

Proof. For $m \geq 1$,

$$\int_m^\infty \log(X/m) X^{-s-1} dX = \frac{m^{-s}}{s^2}.$$

Applying this to (2) gives

$$\int_1^\infty Q_X(j) X^{-s-1} dX = \frac{C_j \zeta(z) + P_j(z)}{s^2}.$$

In (3), substitute $X = kY$; the factor is $k^{-s-1/2}$. Hence the Möbius series contributes $1/\zeta(z)$ and gives (19). On the positive real s -axis, $\zeta(s+1/2)$ has no zero. Its pole at $s = 1/2$ makes $1/\zeta$ vanish, so the apparent point is removable. $\square$

9 Two-row noncancellation

For $j = 2, 3$, (18) simplifies to

$$P_2(z) = 2^{1-z} - 1 - 3^{-z}, \quad (20)$$

$$3P_3(z) = 5 \cdot 3^{-z} - 2^{-z} - 1 - 3 \cdot 4^{-z}. \quad (21)$$

Lemma 9.1 (No common zero). *P_2 and P_3 have no common zero in $\Re z > 0$.*

Proof. Put $a = 2^{-z}$ and $b = 3^{-z}$. If $P_2(z) = 0$, then $b = 2a - 1$. Substitution in (21) gives

$$-3(a-1)(a-2) = 0.$$

For $\Re z > 0$, $|a| = 2^{-\Re z} < 1$, so neither $a = 1$ nor $a = 2$ is possible. $\square$

10 Landau's theorem and the proposed conclusion

We use the classical Mellin form of Landau's theorem: if $f \geq 0$ is locally integrable on $[1, \infty)$ and has finite abscissa of convergence σ_c , then the real point $s = \sigma_c$ is a singularity of its Mellin transform unless the defining integral is holomorphic beyond it.

Theorem 10.1 (Proposed Riemann Hypothesis conclusion). *Theorems 5.3 and 6.1, together with the classical analytic facts used in Lemma 8.1, imply the Riemann Hypothesis.*

Proof. By Theorem 7.1, the functions $X \mapsto c_X(2)$ and $X \mapsto c_X(3)$ are nonnegative. Equation (2) gives the elementary growth bound $c_X(j) = O_j(\sqrt{X} \log(2X))$, so each Mellin abscissa is finite.

Lemma 8.1 gives a continuation analytic at every positive real s . Landau’s theorem therefore forces the defining Mellin integrals to be holomorphic throughout $\Re s > 0$.

Suppose $\zeta(\rho) = 0$ with $\Re \rho > 1/2$ and put $s_\rho = \rho - 1/2$. By Lemma 9.1, at least one of $P_2(\rho)$ and $P_3(\rho)$ is nonzero. The corresponding expression (19) has a nonremovable pole at $s = s_\rho$, contradicting holomorphy of the defining nonnegative Mellin integral there. Thus no zero lies to the right of the critical line. The functional equation reflects the conclusion to the left, so all nontrivial zeros lie on $\Re s = 1/2$. $\square$

11 No-RH-input audit

The proposed proof uses only:

- finite squarefree factorization and parity labels;
- the exact coefficient identities (8)–(10);
- finite scale descent by primes at least 67;
- a finite/directly certified terminal Hall theorem;
- absolutely convergent Dirichlet series in $\Re z > 1$;
- meromorphic continuation and the functional equation of ζ ;
- Landau’s theorem for nonnegative Mellin transforms.

It does not assume RH, GRH, a zero-density estimate, a Mertens bound, a power-saving prime number theorem, a square-root prime error, or an endpoint-deficit inequality.

12 Computer-assisted proof object and reproducibility

The accompanying deterministic archive includes:

- this L^AT_EX source and compiled PDF;
- theorem and claim-status ledgers;
- a lightweight exact checker for the causal coefficients, two-row Mellin algebra, owner/rank fixtures, and mutation firewalls;
- the exact PR/commit/blob lock for the compact and MPFR terminal theorem;
- SHA-256 manifests and a one-command publisher.

The lightweight checker is not substituted for Theorems 5.3 or 6.1; it reports this limitation in its retained JSON.

13 Hostile review checklist

A reviewer should reject the proposal at the first occurrence of:

1. a source atom receiving two first-owner histories;
2. a causal substitution whose coefficients do not reproduce the parent exactly;
3. an unresolved branch whose scale rank does not fall;
4. a terminal leaf with $y \geq 67$ or with a non- P_{61} small divisor;
5. a parity swap lost between source and row observation;
6. a Target–Lorenz interval not covered by the compact or MPFR certificate;
7. an ordinary and radix-four coefficient mismatch inside the terminal realization;
8. an incorrect factor $k^{-s-1/2}$ in the Mellin substitution;
9. a common zero of P_2, P_3 in $\Re z > 0$;
10. a failure of the hypotheses or orientation of Landau’s theorem.

14 Conclusion

The manuscript replaces the refuted cross- n reservoir transport by a finite paired source computation. The proof candidate is unusually concentrated: the arithmetic burden is the exact finite stopping line plus the terminal directed Hall theorem; the analytic finish is a two-row reciprocal-zeta Mellin argument. The paper is submitted as a complete unconditional proposal, not as an accepted resolution.

A Exact coefficient checks

From (1),

$$A_j - B_j = (j + 1)C_j > 0, \quad B_j = 1 - C_j.$$

For the hazard coefficients,

$$\sum_i \alpha_i = \sum_i r_i \lambda_i \leq r_1 \sum_i \lambda_i < 1/\sqrt{67}.$$

For rows two and three, the common-zero elimination in Lemma 9.1 is exact algebra over the two exponential variables 2^{-z} and 3^{-z} .

B Frozen terminal theorem metadata

Item	Frozen value
Projective terminal source PR	#550 at 20646a78c3e8843001cb49ea0c9741f6d0d446f7
Compact theorem	L-91781, blob 2c16327c6653009667fa06ddbb24628ff583c0fe
Proportional reduction	L-91780, blob aeddda0288be0b8f5484d75e5bdee20c2ce4d70f
Tail proof object	ba6b137b64819a9d01e706ffffdf100b091e49034895e6eeb280b56f514c99c5
Packet verification	cc0696b466fbd733043471b322a57f681fd4b845db8cd7225b0861d7a273bb72
MPFR precision	256 bits
Tail events	51,118,080

References

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